

7 (a) ***The root-mean-square (r.m.s.)*** voltage of an alternating current is defined as the **steady direct voltage** which converts electrical energy to other forms of energy in a given resistance **at the same rate as the a.c.** [2]

(b) (i) Turns Ratio [1]

$$\frac{N_s}{N_p} = \frac{V_s}{V_p} = \frac{240 \times 10^3}{12.0 \times 10^3}$$

$$= 20$$

(ii) Transmitting current = $\frac{\text{power}}{\text{voltage}} = \frac{20 \times 10^6}{240 \times 10^3}$

$$= 83.3 \text{ A}$$

Power lost = $I^2 R = (83.3)^2 (200) = 1387778 \text{ W}$

$$= 1390 \text{ kW}$$

peak power = 2 x mean power = 2800 kW

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